DEEDY: A Thai-Centric Multi-Agent Framework for Social-Media Marketing Campaign Analysis

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***Abstract*—This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), a Thai-centric multi-agent framework for analyzing social-media campaign response. DEEDY separates factual campaign evidence from authentic consumer reactions, constructs a grounded simulated world, and combines language-model agents with a scalable stochastic population. The evaluation uses 5,000 comments from five Thai campaigns. Across five experiments, 1,300 generation calls test grounding, Thai prompting, response models, social mechanisms, and message clarity. A separate GLM-5.3-Flash judge evaluates 741 real–synthetic comment pairs twice with reversed presentation order. Authentic comments were preferred in 59.3% of judgments, although synthetic reactions received higher naturalness and campaign-relevance scores. The system therefore demonstrates an auditable workflow and useful sensitivity analysis, but not behavioral fidelity, campaign effectiveness, or deterministic forecasting.**

***Index Terms*—LLM agents, social-media marketing, campaign analysis, social simulation, Thai NLP, hybrid agent-based modeling, key opinion leaders, information operations**

I. Introduction

Social-media campaigns develop through exposure, reply, sharing, and recommendation rather than through isolated comments. Sentiment counts describe a snapshot but not the interaction process that produces it. Agent-based modeling (ABM) provides a bottom-up representation of local decisions and their collective effects [1]; LLM agents add context-dependent language, memory, and open-ended actions [2], [3].

DEEDY adapts the MiroFish/OASIS simulation workflow [4], [5] to campaign analysis. It builds audience actors from approved campaign evidence, simulates exposure and social actions, and compares the resulting discourse with authentic reactions that were hidden during generation. Counterfactual differences are treated as hypotheses, not predictions of a specific person or the Thai market.

Thai campaigns require more than prompt translation. Informal comments mix Thai and English, emoji, non-standard spelling, sarcasm, indirect disagreement, and culturally specific politeness. Thai resources have improved [6], [7], [8], [9], but linguistic form does not guarantee cultural competence [10]. Moreover, plausible generated text does not prove behavioral fidelity [11], [12], [13]. DEEDY therefore separates construction evidence from evaluation reactions and reports uncertainty across repeated conditions.

Our contributions are threefold:

- a provenance-preserving Thai campaign corpus and evidence split;
- a grounded hybrid simulation with configurable exposure, memory, and Thai-language action generation; and
- a leakage-resistant evaluation combining distributional metrics, reversed-order LLM judging, and planned native-Thai review.

II. Related Work

Generative Agents introduced memory, reflection, and planning for believable behavior [2]; S^3, OASIS, and GenSim extended this idea to networked social environments, multiple online actions, and reusable simulations [14], [5], [15]. MiroFish connects a reality seed to graph-grounded personas, social simulation, and report generation [4]. DEEDY adapts this workflow to Thai campaign analysis and adds an explicit separation between evidence used to construct a scenario and evidence used to evaluate it.

This separation is important because plausible language is not equivalent to behavioral validity. Algorithmic fidelity compares generated and human population distributions [11], while conditioned-comment prediction directly compares generated and authentic responses to the same stimulus [13]. Generative ABMs should therefore be validated incrementally before counterfactual claims are made [16].

Thai social-media analysis adds informal spelling, code-mixing, emoji, sarcasm, indirect stance, and culturally specific politeness. Thai NLP tools and pretrained models provide useful linguistic infrastructure [6], [7], [8], [9], but cultural knowledge remains a separate capability [10]. Conversation context is also necessary because sarcasm can reverse literal polarity [17]. These findings motivate evaluation of naturalness, cultural fit, stance, and authenticity rather than sentiment alone.

III. DEEDY System Architecture and Workflow

A. Evidence Preparation

Public campaign posts and comments are collected from Facebook, Instagram, and TikTok through recorded Apify queries [18]. Text is preserved with emoji, Thai–English code-mixing, repeated characters, and thread cues because these may carry pragmatic meaning. Direct identifiers are removed from research views, while campaign, platform, post, time, and collection provenance are retained. The resulting corpus is not treated as a random sample of Thai consumers: platform

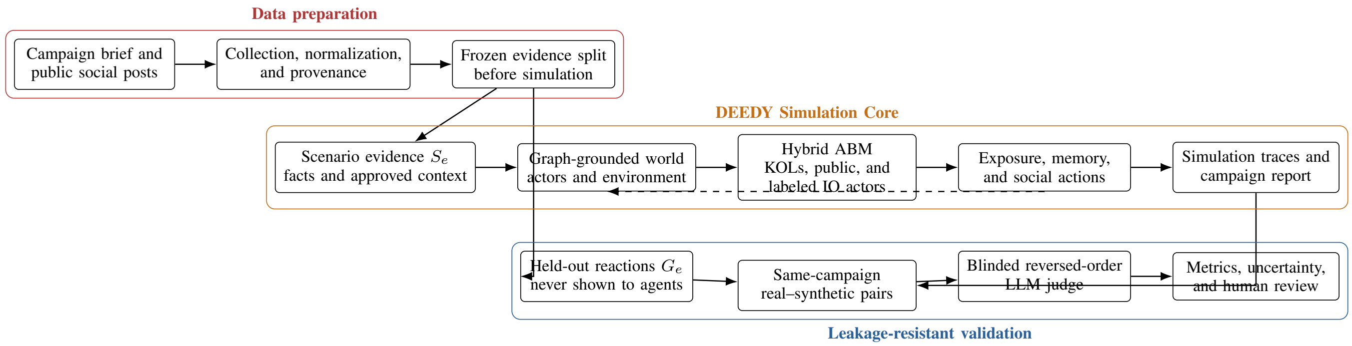

Fig. 1. DEEDY architecture and executed evaluation workflow. Campaign facts construct the simulated world, while authentic reactions remain hidden until outputs are scored. The dashed arrow represents memory feedback within the simulation.

visibility, ranking, moderation, deletion, and source selection can affect which comments are observable.

For campaign e, evidence is partitioned before generation as

$$D_e = S_e \,\dot{\cup}\, G_e, \tag{1}$$

where S_e contains only facts and context permitted to construct the scenario, and G_e contains authentic reactions reserved for evaluation. Comments, their summaries, and derived sentiment proportions from G_e are not supplied to agents. This design prevents the simulator from repeating the answers against which it will later be scored.

B. DEEDY Simulation Core

Figure 1 presents the conceptual architecture. The deployed component is the *DEEDY Simulation Core*. It executes five steps: (1) scenario construction, (2) actor and environment initialization, (3) hybrid-agent decision making, (4) exposure and network propagation, and (5) trace aggregation and report generation. The architecture belongs to the DEEDY application; external simulation systems are discussed only as prior work and implementation foundations.

1) Scenario, Actors, and Memory: Campaign facts are organized as entities, relations, and retrievable scenario memories. This graph-grounding step is retained because an agent must recover the correct product, campaign condition, creator, and prior event when deciding how to react; it is not used to inject authentic evaluation comments. Audience roles are sampled with interests, prior attitude, activity, language register, Thai–English mixing, and uncertainty. Evidence-backed attributes remain separate from sampled attributes, and unsupported details remain assumptions. The environment defines the platform action space, time step, feed capacity, and exposure policy.

Each cognitive agent maintains short-term interaction memory and longer-term scenario memory. At a simulation step, the agent retrieves relevant facts, observes a bounded feed and recent interaction context, updates its private stance, and selects like, share, comment, reply, or silence. Thai generation preserves code-mixing and uncertainty without forcing dialect caricatures. Thread context is available for replies because Thai stance and sarcasm may depend on earlier turns [17].

2) Hybrid Agent-Based Modeling (Hybrid ABM): DEEDY partitions the simulated population as $\mathcal{A} = \mathcal{K} \,\dot{\cup}\, \mathcal{P} \,\dot{\cup}\, \mathcal{I}$. The cognitive KOL tier $\mathcal{K}$ contains a small configurable set of influential or analytically important actors. These agents retrieve relevant memories, observe a bounded feed, update private stance, and generate a context-aware Thai action or remain silent. They represent key opinion-leader roles inside the scenario, not claims about the behavior of named real people.

The public tier $\mathcal{P}$ is larger and stochastic. Each public agent retains an audience profile, prior attitude, exposure history, and network position, but uses a probabilistic response rule to preserve population scale. For an exposed public agent i at time t,

$$\Pr(a_{i,t} \neq \varnothing) = \sigma(\beta_0 + \beta_b b_i + \boldsymbol{\beta}_d^\top \mathbf{d}_i + \beta_s s_{t-1} + \beta_x x_{i,t}). \tag{2}$$

Here b_i is prior attitude, $\mathbf{d}_i$ contains sampled audience attributes, s_{t-1} is the preceding social signal, and $x_{i,t}$ summarizes exposure and neighbors. A categorical draw then selects like, share, comment, reply, or silence. Tier-specific results are retained so KOL output is not silently treated as public opinion.

The optional astroturfing and information-operations (IO) tier $\mathcal{I}$ injects a pre-registered proportion of coordinated positive promotion or negative attack. Every injected actor and action is labeled and reported separately from organic KOL and public activity. This tier supports counterfactual robustness testing as injection rate, timing, message consistency, and amplification vary; it is not a detector of real accounts and does not assert that authentic Thai users follow coordinated actors.

Initial exposure can be chronological, interest-based, popularity-based, For-You-Page seeding, or direct KOL injection. Later shares propagate through the configured network. Feed rules, KOL selection, graph topology, and IO prevalence are experimental assumptions rather than reconstructions of a platform's proprietary recommender.

3) Outputs and Validation Path: Every output connects an action to its actor, exposure, retrieved evidence, parent action, and simulated time. The simulation report summarizes sentiment, stance, narratives, reach, engagement, influential actors, and disagreement across repeated runs. Separately, visible generated reactions are paired with same-campaign authentic comments from G_e for blinded evaluation. Aggregate reference distributions are loaded only after generation and judging. This traceable separation supports error analysis and counterfactual comparison but does not make a simulated difference a real-world causal effect.

IV. Evaluation and Experimental Results

A. Data and Experimental Design

The evaluation covers five Thai campaigns: Parameter Gelato, KFC Bucket Ware, MK Buffet, Walk/Run for Glasses by Top Charoen, and Thai Chuai Thai Plus. The corpus contains 5,000 authentic comments from 33 posts, with 1,000 comments per campaign. Aggregate sentiment analyses were used only after generation. The available crawl contained no reply links, so observed cascade correspondence could not be evaluated; network results are reported only as model sensitivity.

Experiments 1–3 test grounding, Thai prompting, and the response model. Experiment 4 varies feed and network assumptions and generates agent reactions. Experiment 5 compares a baseline campaign packet with a clarity frame that makes conditions and uncertainties more explicit. Ten random seeds are used per campaign-condition combination. Jensen–Shannon distance (JSD) measures sentiment-distribution divergence, while text quality is assessed with diversity, semantic correspondence [19], and blinded judgment. Tables II and III report the principal numeric results.

Table I makes the Experiment 4–5 inputs and scoring boundary explicit. Agents received the scenario facts and stated uncertainties, but not the authentic sentiment proportions in the final column. Those proportions were computed from real comments and loaded only after generation as a weak aggregate reference rather than a human gold standard.

Across Experiments 1–5, 1,300 generation calls completed. Experiments 4–5 produced 6,400 agent records from 800 calls. Of these, 741 calls contained at least one visible reaction; 59 all-silence calls were retained in action-rate analysis but excluded from text comparison. Experiment 1 did not establish a grounding advantage, Experiment 2 did not establish a Thai-context advantage, and Experiment 3 did not establish superiority of either LLM or the stochastic baseline. High Thai-character ratios were not interpreted as naturalness or cultural competence.

Experiment 4 changed LLM outputs only modestly across prompt-level feed/network descriptions and under-produced negative reactions. Table IV exposes the six conditions behind the ranges in Table III. DeepSeek's lowest mean JSD occurred under the chronological/matched-random description (0.2995), whereas Qwen's occurred under popularity/matched-random (0.3274). Negative share never exceeded 0.0225, despite authentic campaign references ranging from 0.049 to 0.362.

To isolate the social mechanism from language generation, a separate sweep used 500 public agents under each feed–network combination. Table V shows that the graph assumption changed participation and diffusion much more than the ranking policy: the matched-random graph reached almost the entire population, whereas the post-affiliation proxy reached less than half.

Ranking changed JSD and polarization within each graph, but did not overcome the reach gap between graphs. Because the corpus did not preserve reply edges, neither topology represents an observed Thai interaction network. These results are sensitivity bounds, not evidence that a platform used one of the tested mechanisms.

For Experiment 5, clarity improved DeepSeek JSD by 0.0075 but worsened Qwen JSD by 0.0014. Across 100 matched generator–campaign–seed pairs, the clarity-minus-baseline change was -0.0030 (95% empirical interval -0.1888–0.1261). The interval spans zero and does not establish a message-clarity advantage.

The generative panels measure one-step agent decisions and reaction text under controlled scenario descriptions. They do not constitute a temporally ordered reply cascade: feed and network conditions enter the cognitive-agent context, while the non-generative sweep separately tests diffusion. This distinction prevents comment-level realism scores from being misread as validation of the full network trajectory.

Table VI reports the paired uncertainty estimates that support these restrained conclusions. Differences are computed within matched campaign–seed units, so negative values favor the first-listed condition when lower JSD is better. Every interval crosses zero; small differences in sample means should therefore not be promoted to treatment or model effects.

B. Blinded LLM-as-a-Judge Evaluation

Each eligible synthetic reaction was paired with a distinct authentic comment from the same campaign. Direct identifiers and contact patterns were removed; the judge received only the campaign facts and two unlabeled texts. Each of the 741 pairs was evaluated twice with A/B order reversed, producing 1,482 valid judgments. GLM-5.3-Flash scored Thai naturalness, campaign relevance, social-media plausibility, pragmatic/cultural fit, contextual specificity, and unsupported-claim risk on five-point scales. It also predicted sentiment and selected which text appeared more authentic. Pair reversal was used because LLM judges may exhibit presentation and position biases [20]. Aggregate sentiment references were hidden until all judgments were complete.

The authentic comment was preferred in 59.3% of judgments, the synthetic reaction in 40.6%, and a tie in 0.1%. Treating a tie as half a win gives a synthetic-realism score of 0.406 (bootstrap 95% CI 0.374–0.438), below the 0.5 indifference point. Reversed-order preference agreed for 83.5% of pairs, while candidate A was selected in 55.9% of non-ties, confirming measurable position sensitivity.

Table VII reveals a useful tension. Synthetic reactions were rated more natural, relevant, culturally appropriate, and specific, yet authentic comments were more often selected as real. Social plausibility and unsupported-claim risk showed no clear difference. The judge-labeled sentiment distribution diverged less from the aggregate reference for authentic comments (JSD = 0.077) than for synthetic reactions (0.215), reinforcing the under-production of negative response.

TABLE I
FROZEN EXPERIMENT 4–5 CAMPAIGN SCENARIOS AND AUTHENTIC SCORING-ONLY SENTIMENT REFERENCES. VALUES ARE NEGATIVE/NEUTRAL/POSITIVE SHARES AND WERE NOT INCLUDED IN AGENT PROMPTS.

Campaign	Scenario facts available to simulated agents	Reference
Parameter Gelato	Premium six-flavour gelato launch at Siam Paragon; price, serving size, and availability left uncertain.	0.362/0.476/0.162
KFC Bucket Ware	Limited reusable container: THB 299 qualifying purchase plus THB 99 redemption; dates, branches, and stock conditional.	0.049/0.309/0.642
MK Buffet	THB 299 net, 33 items, four refillable seafood items, and optional THB 59/129 upgrades; branch availability conditional.	0.136/0.444/0.420
Top Charoen	Record a chosen walk/run distance and redeem eligible eyewear; tracking, entitlement, charges, stock, and fulfilment conditional.	0.050/0.700/0.250
Thai Chuai Thai Plus	Government 60/40 co-payment through Paotang G-Wallet; eligibility, allowance, merchants, dates, and exclusions follow official rules.	0.175/0.275/0.550

TABLE II
EXPERIMENTS 1–3: MEAN SENTIMENT JSD OVER FIVE CAMPAIGNS AND TEN SEEDS. THE AGGREGATE REFERENCE IS A DIAGNOSTIC CALIBRATION TARGET, NOT AN INDEPENDENT HOLDOUT. NO REPORTED PAIRED INTERVAL EXCLUDED ZERO.

Experiment	Condition	DeepSeek	Qwen	Stochastic ABM
Exp. 1	Grounded scenario	0.2572	0.2859	–
Exp. 1	Ungrounded prompt	0.2635	0.2388	–
Exp. 2	Thai-context prompt	0.2572	0.2859	–
Exp. 2	Translated/unadapted	0.2604	0.2523	–
Exp. 2	Normalized, no context	0.2460	0.2548	–
Exp. 3	Response-model comparison	0.2572	0.2859	0.2779

TABLE III
EXPERIMENTS 4–5 LLM REACTION RESULTS. EXPERIMENT 4 ENTRIES ARE RANGES OVER SIX FEED/NETWORK DESCRIPTIONS; EXPERIMENT 5 ENTRIES AVERAGE FIVE CAMPAIGNS AND TEN SEEDS. VISIBLE ACTION EXCLUDES SILENCE.

Experiment	Generator	Condition	Runs	Sentiment JSD	Visible action	Negative share
Exp. 4	DeepSeek	Feed/network range	300	0.2995–0.3233	0.760–0.783	0.008–0.023
Exp. 4	Qwen	Feed/network range	300	0.3274–0.3432	0.740–0.753	0.000–0.005
Exp. 5	DeepSeek	Baseline	50	0.3189	0.7775	0.0100
Exp. 5	DeepSeek	Clarity	50	0.3114	0.8200	0.0250
Exp. 5	Qwen	Baseline	50	0.3335	0.7450	0.0000
Exp. 5	Qwen	Clarity	50	0.3349	0.7500	0.0050

Table VIII shows that DeepSeek was closer to the indifference point than Qwen in both experiments, but this is a comparison under one judge rather than evidence of general model superiority. The unequal sample counts also matter: all-silence calls have no synthetic text to pair and are therefore excluded from the realism comparison while remaining in action- rate analysis. KFC was the only campaign with a synthetic-realism score slightly above 0.5, whereas Top Charoen had the largest authenticity gap. Campaign slices are diagnostic because sample eligibility and judge priors may differ.

C. Campaign-Level Results

Table IX reports all five campaign slices. No campaign's realism interval lay wholly above the 0.5 indifference point. KFC was closest to indifference, while Top Charoen was most readily distinguished from authentic discussion. Sentiment correspondence did not always move with authenticity: Top Charoen's synthetic sentiment distribution was close to the aggregate reference even though its generated comments were rarely preferred as real. Distribution matching and comment-level realism therefore measure different properties and should be reported together.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

The results do not simply make DEEDY appear more accurate. They expose a "too polished and too relevant" failure mode: prompted agents produce coherent standalone responses, whereas authentic threads contain fragments, tagging, repetition, jokes, and side conversations. Optimizing only rubric scores could therefore reduce realism. Experiment 4 evaluates sensitivity to specified feed and network conditions, not reconstruction of platform exposure. Experiment 5 evaluates reactions to hypothetical framing, not a causal effect on consumers. The simulations are consequently most useful for scenario exploration and failure discovery.

The hybrid architecture also clarifies the coverage of the present results. The generative Experiment 4–5 panels exercise the cognitive, KOL-like decision path, while the non-generative mechanism sweep exercises the scaled public tier; they are not yet one end-to-end mixed-population trajectory. The IO tier remained disabled during real–synthetic correspondence because injected coordination would change the estimand and contaminate comparison with organic comments. A proper IO experiment must pre-register injection share, timing, polarity, and amplification; preserve labels on every injected trace; and compare reach, polarization, cross-group adoption, and recovery

TABLE IV
EXPERIMENT 4 LLM COGNITIVE-AGENT SENSITIVITY. EACH ROW AVERAGES FIVE CAMPAIGNS AND TEN SEEDS. MR DENOTES MATCHED RANDOM AND PA THE POST-AFFILIATION PROXY; VISIBLE ACTION EXCLUDES SILENCE.

Generator	Feed/network	Runs	Sentiment JSD	Visible action	Neutral share	Negative share
DeepSeek	Chronological/MR	50	0.2995	0.7825	0.5700	0.0200
DeepSeek	Chronological/PA	50	0.3211	0.7800	0.6050	0.0075
DeepSeek	Interest/MR	50	0.3159	0.7675	0.5800	0.0225
DeepSeek	Interest/PA	50	0.3154	0.7750	0.5925	0.0150
DeepSeek	Popularity/MR	50	0.3051	0.7825	0.5825	0.0075
DeepSeek	Popularity/PA	50	0.3233	0.7600	0.6100	0.0075
Qwen	Chronological/MR	50	0.3291	0.7450	0.6400	0.0025
Qwen	Chronological/PA	50	0.3277	0.7450	0.6500	0.0050
Qwen	Interest/MR	50	0.3389	0.7400	0.6625	0.0000
Qwen	Interest/PA	50	0.3432	0.7400	0.6725	0.0025
Qwen	Popularity/MR	50	0.3274	0.7475	0.6425	0.0000
Qwen	Popularity/PA	50	0.3431	0.7525	0.6550	0.0000

TABLE V
EXPERIMENT 4 NON-GENERATIVE SOCIAL-MECHANISM SENSITIVITY ($N = 500$; MEANS OVER FIVE CAMPAIGNS AND TEN SEEDS). JSD IS A DIAGNOSTIC COMPARISON WITH THE AGGREGATE SENTIMENT REFERENCE; CASCADE SIZE IS A FRACTION OF AGENTS.

Feed policy	Network assumption	Action rate	Reach	JSD	Polarization	Largest cascade
Chronological	Matched random	0.968	0.997	0.077	0.172	0.251
Interest	Matched random	0.970	0.997	0.102	0.138	0.224
Popularity	Matched random	0.968	0.997	0.120	0.126	0.239
Chronological	Post-affiliation proxy	0.402	0.463	0.082	0.169	0.159
Interest	Post-affiliation proxy	0.400	0.463	0.097	0.163	0.161
Popularity	Post-affiliation proxy	0.400	0.461	0.115	0.164	0.159

TABLE VI
PAIRED SENTIMENT-JSD CONTRASTS ACROSS EXPERIMENTS 1–3 AND 5. INTERVALS ARE EMPIRICAL 2.5–97.5 PERCENTILES OVER MATCHED CAMPAIGN–SEED UNITS. LOWER JSD IS BETTER.

Experiment	Generator/comparison	Contrast (first minus second)	Mean	95% interval
Exp. 1	DeepSeek	Grounded − ungrounded	−0.0063	[−0.1483, 0.1756]
Exp. 1	Qwen	Grounded − ungrounded	0.0471	[−0.3313, 0.3484]
Exp. 2	DeepSeek	Thai-context − translated	−0.0032	[−0.1444, 0.1344]
Exp. 2	Qwen	Thai-context − translated	0.0337	[−0.1988, 0.3302]
Exp. 3	Cross-model	DeepSeek − Qwen	−0.0287	[−0.3592, 0.2179]
Exp. 3	Cross-tier	DeepSeek − stochastic ABM	−0.0207	[−0.2341, 0.1955]
Exp. 3	Cross-tier	Qwen − stochastic ABM	0.0081	[−0.3765, 0.4340]
Exp. 5	Pooled generators	Clarity − baseline	−0.0030	[−0.1888, 0.1261]

TABLE VII
BLINDED JUDGE RESULTS OVER 741 REAL–SYNTHETIC PAIRS. SCORE GAPS ARE SYNTHETIC MINUS REAL; RISK IS THE ONLY DIMENSION WHERE HIGHER IS WORSE.

Dimension	Real	Synthetic	Gap (95% CI)
Thai naturalness	3.77	4.19	0.42 [0.29, 0.57]
Campaign relevance	2.83	4.46	1.64 [1.51, 1.77]
Social plausibility	3.84	3.88	0.04 [−0.11, 0.20]
Cultural fit	3.57	4.07	0.50 [0.35, 0.65]
Context specificity	2.23	3.14	0.91 [0.76, 1.05]
Unsupported-claim risk	1.83	1.87	0.04 [−0.06, 0.14]

with a same-seed no-IO condition. It is therefore an auditable stress-test capability, not reported evidence of real astroturfing.

The study has four principal limitations. First, manually selected public posts form a convenience sample rather than a representative Thai population. Second, missing reply edges prevent validation of propagation and cascade structure. Third, the sentiment reference is an aggregate weak annotation, not an independent human gold standard. Fourth, a single external LLM judge can share model priors with the generators and showed position sensitivity. Sanitization reduces but cannot eliminate privacy risk when authentic text is sent to an external service. Future evaluation should use a pre-registered campaign-level holdout, reply- and time-preserving traces, a Thai-specialized generator, multiple judges, and blinded native-Thai raters with reported agreement.

DEEDY contributes a leakage-resistant architecture for connecting grounded Thai campaign scenarios, hybrid multi-agent simulation, and authentic-response evaluation. The completed pilot demonstrates end-to-end execution and reveals where the model departs from observed discourse; it does not establish market forecasting, campaign effectiveness, or behavioral fidelity. This distinction is central to using generative simulation as an auditable hypothesis tool rather than a substitute for field research.

REFERENCES

[1] C. M. Macal and M. J. North, "Tutorial on agent-based modelling and simulation," in *Proceedings of the 2010 Winter Simulation Conference*, 2010, pp. 151–162.

TABLE VIII
BLINDED JUDGE RESULTS BY EXPERIMENT AND GENERATOR. REALISM EQUALS SYNTHETIC PREFERENCE PLUS HALF THE TIE SHARE. SCORE GAPS ARE SYNTHETIC MINUS REAL; POSITIVE VALUES FAVOR THE SYNTHETIC TEXT.

Experiment	Generator	Pairs	Real preferred	Synthetic preferred	Realism (95% CI)	Thai gap	Plausibility gap
Exp. 4	DeepSeek	300	0.5533	0.4467	0.4467 [0.4000, 0.5000]	0.5200	0.1917
Exp. 4	Qwen	260	0.6481	0.3500	0.3510 [0.2990, 0.4038]	0.2942	−0.1731
Exp. 5	DeepSeek	100	0.5400	0.4600	0.4600 [0.3749, 0.5450]	0.7200	0.4100
Exp. 5	Qwen	81	0.6296	0.3642	0.3673 [0.2747, 0.4630]	0.0926	−0.2531

TABLE IX
CAMPAIGN-LEVEL JUDGE RESULTS. REALISM EQUALS SYNTHETIC PREFERENCE PLUS HALF THE TIE SHARE. JSD IS SHOWN AS AUTHENTIC/SYNTHETIC; LOWER IS BETTER.

Campaign	Pairs	Realism (95% CI)	JSD (R/S)
Parameter Gelato	155	0.431 [0.363, 0.498]	0.181/0.428
KFC Bucket Ware	142	0.511 [0.440, 0.585]	0.288/0.274
MK Buffet	149	0.450 [0.376, 0.527]	0.123/0.220
Top Charoen	155	0.247 [0.182, 0.313]	0.068/0.049
Thai Chuai Thai Plus	140	0.404 [0.332, 0.475]	0.174/0.376

[2] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–22.

[3] J. S. Park, L. Popowski, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Social simulacra: Creating populated prototypes for social computing systems," in *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 2022, pp. 1–18.

[4] MiroFish Contributors, "MiroFish: A simple and universal swarm intelligence engine," Open-source software repository, 2026, accessed 12 August 2026. [Online]. Available: https://github.com/666ghj/MiroFish

[5] Z. Yang, Z. Zhang, Z. Zheng, Y. Jiang, Z. Gan, Z. Wang, Z. Ling, J. Chen, M. Ma, B. Dong, P. Gupta, S. Hu, Z. Yin, G. Li, X. Jia, L. Wang, B. Ghanem, H. Lu, C. Lu, W. Ouyang, Y. Qiao, P. Torr, and J. Shao, "OASIS: Open agent social interaction simulations with one million agents," in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: https://arxiv.org/abs/2411.11581

[6] R. Arreerard, S. Mander, and S. Piao, "Survey on Thai NLP language resources and tools," in *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. European Language Resources Association, 2022, pp. 6495–6505. [Online]. Available: https://aclanthology.org/2022.lrec-1.697/

[7] W. Phatthiyaphaibun, K. Chaovavanich, C. Polpanumas, A. Suriyawongkul, L. Lowphansirikul, P. Chormai, P. Limkonchotiwat, T. Suntorntip, and C. Udomcharoenchaikit, "PyThaiNLP: Thai natural language processing in python," in *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software*, 2023, pp. 25–36. [Online]. Available: https://aclanthology.org/2023.nlposs-1.4/

[8] L. Lowphansirikul, C. Polpanumas, A. Rutherford, and S. Nutanong, "WangchanBERTa: Pretraining transformer-based thai language models," 2021.

[9] K. Pipatanakul, P. Manakul, N. Nitarach, W. Sirichotedumrong, S. Nonesung, T. Jaknamon, P. Pengpun, P. Taveekitworachai, A. Na-Thalang, S. Sripaisarnmongkol, K. Jirayoot, and K. Tharnpipitchai, "Typhoon 2: A family of open text and multimodal thai large language models," 2024.

[10] D. Kim, S. Lee, Y. Kim, A. Rutherford, and C. Park, "Representing the under-represented: Cultural and core capability benchmarks for developing thai large language models," in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 4114–4129. [Online]. Available: https://aclanthology.org/2025.coling-main.278/

[11] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate, "Out of one, many: Using language models to simulate human samples," *Political Analysis*, vol. 31, no. 3, pp. 337–351, 2023.

[12] J. Mooney, J. Woldense, Z. R. Jia, S. A. Hayati, M. H. Nguyen, V. Raheja, and D. Kang, "Are LLM agents behaviorally coherent? latent profiles for social simulation," 2025.

[13] N. Schwager, S. Münker, A. Plum, and A. Rettinger, "Towards simulating social media users with LLMs: Evaluating the operational validity of conditioned comment prediction," in *Proceedings of the 15th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 2026, pp. 208–221. [Online]. Available: https://aclanthology.org/2026.wassa-1.16/

[14] C. Gao, X. Lan, Z. Lu, J. Mao, J. Piao, H. Wang, D. Jin, and Y. Li, "S^3: Social-Network Simulation System with Large Language Model-Empowered Agents," 2023.

[15] J. Tang, H. Gao, X. Pan, L. Wang, H. Tan, D. Gao, Y. Chen, X. Chen, Y. Lin, Y. Li, B. Ding, J. Zhou, J. Wang, and J.-R. Wen, "GenSim: A general social simulation platform with large language model based agents," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, 2025, pp. 143–150. [Online]. Available: https://aclanthology.org/2025.naacl-demo.15/

[16] L. Cross, N. Haber, and D. L. K. Yamins, "Validating generative agent-based models of social norm enforcement: From replication to novel predictions," 2025.

[17] D. Hazarika, S. Poria, S. Gorantla, E. Cambria, R. Zimmermann, and R. Mihalcea, "CASCADE: Contextual sarcasm detection in online discussion forums," in *Proceedings of the 27th International Conference on Computational Linguistics*, 2018, pp. 1837–1848. [Online]. Available: https://aclanthology.org/C18-1156/

[18] Apify, "Actors documentation: Runs and builds," Online documentation, 2026, accessed 12 August 2026. [Online]. Available: https://docs.apify.com/actors/running/runs-and-builds

[19] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=SkeHuCVFDr

[20] L. Shi, C. Ma, W. Liang, W. Ma, and S. Vosoughi, "Judging the judges: A systematic study of position bias in LLM-as-a-judge," 2024.